DAY 13 | PRE-READ

# Checking Your Work and Moving Data

A light read before Day 13. About 15 minutes, no code required.

Prepared by **Ahmed** | linkedin.com/in/ahmed-al-imaad

## Why read this?

Tomorrow has two halves. In the morning you learn to write tests: small programs that check your programs. In the afternoon you step into data engineering: how raw data travels from where it is created to where it helps people decide.

## 1. Tests are like a pilot's checklist

Before every flight, pilots run through a checklist even though they have flown thousands of times. It takes minutes and prevents disasters. Automated tests are a checklist for code. Every time you change something, the computer re-checks everything in seconds.

> **Key idea:** tests give you the confidence to change code without fear of breaking what already works.

## 2. Three sizes of test

| Test size | Everyday example |
|---|---|
| Small (unit) | Checking that one light switch works. |
| Medium (integration) | Checking that the switch, wiring and bulb work together. |
| Large (end to end) | Walking into the room at night and seeing the light come on. |

## 3. The journey of data

Think of water. It falls as rain (data is generated by apps and devices), flows into reservoirs (data is collected, or **ingested**), is stored, then cleaned at a treatment plant (data is **processed**), and finally arrives at your tap (dashboards, reports and AI models). Data engineers build and maintain the pipes.

## 4. Clean first, or store first?

There are two common approaches. **ETL** cleans data before storing it, like washing vegetables before putting them in the fridge. **ELT** stores everything first and cleans it later inside a powerful database, like putting the shopping away quickly and preparing it when you cook. Both are valid;

modern cloud platforms often prefer ELT.

## 5. Think about these before class

- What checks do you do before submitting an important document at work?
- Name three places where data about you is generated every day.
- What could go wrong if the same sales record were loaded into a report twice?

*Anshin (安心): peace of mind*